

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_random_replay.
- Best TSPLIB ATSP holdout gap: atsp_random_replay.
- Best synthetic holdout gap: atsp_failure_replay.

Run Metadata

- run_name: run_20260513_165255_f
- started_at_local: 2026-05-13 16:52:55
- finished_at_local: 2026-05-13 17:03:40
- duration_hhmm: 00:11
- duration_seconds: 644.84
- seed_offset: 5000
- replicate_label: f
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.192059	0.030781	0.11142	0.0	0.78	0.0
atsp_random_replay	random	score_only	False	0.160422	0.027584	0.094003	0.653594	0.78	0.055151
atsp_failure_replay	failure	score_only	False	0.191028	0.009004	0.100016	0.744019	0.58	0.021662
atsp_failure_replay_compression	failure	novelty_gate	True	0.172128	0.027465	0.099796	0.698911	0.78	0.034411

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.192059, synthetic 0.030781, combined 0.11142.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192059 across 4 instances; family means: ft=0.280377, ftv=0.295102, p=0.01726, ry=0.175496.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1887, "family": "ftv", "name": "ftv35", "optimality_gap": 0.281059}, {"best_known_cost": 1286, "cost": 1591, "family": "ftv", "name": "ftv33", "optimality_gap": 0.23717}, {"best_known_cost": 1530, "cost": 1842, "family": "ftv", "name": "ftv38", "optimality_gap": 0.203922}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.160422, synthetic 0.027584, combined 0.094003.
- Accepted-epoch count 3, mean accepted code novelty 0.653594, and final complexity 0.78.
- Adaptation efficiency 0.055151 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.160422 across 4 instances; family means: ft=0.246054, ftv=0.295102, p=0.003737, ry=0.096797.
- Panel synthetic_holdout mean gap 0.027584 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.024723, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1572, "family": "ftv", "name": "ftv33", "optimality_gap": 0.222395}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}, {"best_known_cost": 1530, "cost": 1761, "family": "ftv", "name": "ftv38", "optimality_gap": 0.15098}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.191028, synthetic 0.009004, combined 0.100016.
- Accepted-epoch count 3, mean accepted code novelty 0.744019, and final complexity 0.58.
- Adaptation efficiency 0.021662 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.191028 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.172128, synthetic 0.027465, combined 0.099796.
- Accepted-epoch count 2, mean accepted code novelty 0.698911, and final complexity 0.78.

- Adaptation efficiency 0.034411 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.172128 across 4 instances; family means: ft=0.292976, ftv=0.294482, p=0.001068, ry=0.099986.
- Panel synthetic_holdout mean gap 0.027465 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033248, wind_clusters=0.076613.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1995, "family": "ftv", "name": "ftv38", "optimality_gap": 0.303922}, {"best_known_cost": 1473, "cost": 1748, "family": "ftv", "name": "ftv35", "optimality_gap": 0.186694}, {"best_known_cost": 1286, "cost": 1366, "family": "ftv", "name": "ftv33", "optimality_gap": 0.062208}]

Judge Appendix

Summary of Replay-Aware ATSP Benchmark Results

Condition	Replay Mode	Synthetic Holdout Gap	Held-out TSPLIB Gap	Transfer Gap	Code Novelty (mean)	Complexity	Notes on Transfer vs Novelty
atsp_no_replay	None	0.0308	0.1921	0.1114	0.0	0.78	Baseline. No code novelty. Transfer gaps are highest here.
atsp_random_replay	Random Replay	0.0276	**0.1604**	**0.0940**	0.654	0.78	Best held-out TSPLIB gap and transfer gap, despite only moderate complexity. Code novelty is substantially higher than no replay.
atsp_failure_replay	Failure Replay	**0.0090**	0.1910	0.1000	0.744	0.58	Best synthetic gap with lower complexity, but no TSPLIB gap improvement vs no replay. Transfer gap improved vs no replay. Code novelty increases notably.
atsp_failure_replay_compression	Failure Replay + Compression	0.0275	0.1721	0.0998	0.699	0.78	Compression pressure active. TSPLIB and transfer gaps intermediate. Code novelty higher than no replay but lower than random replay failure replay.

Interpretation Focused on Optimality Gap & Transfer (Main Evidence)

- ****Transfer performance (held-out TSPLIB & transfer gaps) favors atsp_random_replay over other conditions****, with a mean held-out TSPLIB gap of 0.1604 and transfer gap 0.0940, indicating better generalization to unseen ATSP instances.
- Although the ****atsp_failure_replay** condition has the best synthetic holdout gap (0.0090), its held-out TSPLIB gap (0.1910) is comparable to no replay (0.1921) and worse than random replay**, so transfer to real-world instances is not improved here.
- Compression-aware failure replay gives ****no significant TSPLIB or transfer gap advantage compared to random or simple failure replay****, despite compression pressure and moderate code novelty.
- ****No replay has the worst transfer performance and zero code novelty****, confirming no replay leads to poorer adaptation to new instances.

Code Novelty vs Transfer

- **Code novelty increases from no replay (0.0) to random replay (~0.65) and failure replay (~0.74), but transfer gap is best for random replay.**
- Failure replay modes (with or without compression) have higher code novelty but do **not improve TSPLIB transfer over random replay**, showing that increased lexical novelty does not guarantee better transfer.
- Since complexity remains relatively stable except for lower in failure replay (0.58), improvements in transfer (random replay) are not simply due to lower model complexity or oversimplification.

Replay Mode Impact

- **No replay results in poorer transfer and zero adaptation efficiency.**
- **Random replay is associated with better held-out TSPLIB and transfer gaps, moderately high code novelty, and stable complexity.**
- **Failure replay improves synthetic gap and induces higher code novelty and reduced complexity, but does not improve TSPLIB transfer or reduce transfer gap as effectively as random replay.**
- **Compression-aware failure replay does not outperform pure failure replay or random replay in transfer metrics, despite added compression pressure and high code novelty.**

Conservative Conclusion

- The **random replay condition provides the best overall transfer to held-out TSPLIB ATSP instances and synthetic benchmarks, with a reasonable balance of complexity and code novelty.**
- Failure replay (with or without compression) boosts code novelty and synthetic performance but does not translate into improved transfer to TSPLIB ATSP benchmark instances.
- Code novelty increases with replay modes, but **lexical novelty alone is not indicative of transfer improvement**, especially as failure replay conditions have higher novelty but not better TSPLIB transfer compared to random replay.
- Compression-aware replay adds little transfer benefit beyond failure replay, suggesting compression pressure may not directly enhance generalization in this context.

Key Metrics Table (condensed)

Replay Mode	Synthetic Gap	Held-out TSPLIB Gap	Transfer Gap	Mean Code Novelty	Complexity	Adaptation Efficiency
No Replay	0.03078	0.19206	0.11142	0.0	0.78	0.0
Random Replay	0.02758	0.16042	0.09400	0.654	0.78	0.055

Replay Mode	Synthetic Gap	Held-out TSPLIB Gap	Transfer Gap	Mean Code Novelty	Complexity	Adaptation Efficiency
Failure Replay	**0.00900* *	0.19103	0.10002	0.744	0.58	0.022
Failure Replay + Compression	0.02747	0.17213	0.09980	0.699	0.78	0.034

Final Notes

- Prioritize random replay as the most effective transfer condition.
- Failure replay improves synthetic gaps and code novelty but not transfer to held-out TSPLIB.
- Compression-aware replay adds complexity management but no transfer advantage.
- Avoid conflating code novelty or lexical novelty with algorithmic invention without supporting transfer gap improvement.